Average-Based Proportional Distribution

Formula

Definitions

- (S) = set of selected services
- (n) = number of services
- (price(s)) = original price of service (s)
- (P) = package price

Step 1: Average service price

$$
AVG = \frac{\sum_{s \in S} price(s)}{n}
$$

Step 2: Split services

Low-priced services:

$$
L = { s \in S \mid price(s) \le AVG }
$$

High-priced services:

$$
H = { s \in S \mid price(s) > AVG }
$$

Step 3: Remaining amount after paying low services

$$
R = P - \sum_{s \in L} price(s)
$$

Step 4: Proportional distribution among high services

$$
HighTotal = \sum_{h \in H} price(h)
$$

For each $h \in H$:

$$
Payout(h) = R \times \frac{price(h)}{HighTotal}
$$

Question 1

A salon offers a package priced at ₹1000.
The following services are included:

Service	Price (₹)
Hair Cut	120
Shaving	80
Hair Colour	600
Facial	1100

Using the average-based proportional distribution method, calculate the payout for each service.

Answer 1

Step 1: Average service price

$$
AVG = \frac{120 + 80 + 600 + 1100}{4}
= \frac{1900}{4}
= 475
$$

Step 2: Split services

- Low services ((\le AVG)): Hair Cut (120), Shaving (80)
- High services ((> AVG)): Hair Colour (600), Facial (1100)

Step 3: Pay low services fully

[
LowTotal = 120 + 80 = 200
]

[
R = 1000 - 200 = 800
]

Step 4: Proportional reduction

[
HighTotal = 600 + 1100 = 1700
]

Hair Colour payout:

[
800 \times \frac{600}{1700} = 282
]

Facial payout:

[
800 \times \frac{1100}{1700} = 518
]

Final payout

Service	Amount (₹)
Hair Cut	120
Shaving	80
Hair Colour	282
Facial	518

Service	Amount (₹)
Total	**1000**

Question 2

A salon offers a package priced at ₹1000.
The following services are included:

Service	Price (₹)
Hair Cut	120
Shaving	80
Hair Colour	600
D-Tan Pack	250

Using the average-based proportional distribution method, calculate the payout for each service.

Answer 2

Step 1: Average service price

[
AVG = \frac{120 + 80 + 600 + 250}{4}
= \frac{1050}{4}
= 262.5
]

Step 2: Split services

- Low services: Hair Cut (120), Shaving (80), D-Tan (250)
- High services: Hair Colour (600)

Step 3: Pay low services fully

[
LowTotal = 120 + 80 + 250 = 450
]

[
R = 1000 - 450 = 550
]

Step 4: Proportional reduction

Only one service is above average:

[
HairColourPayout = 550
]

Final payout

Service	Amount (₹)
Hair Cut	120
Shaving	80
D-Tan	250
Hair Colour	550
Total	**1000**

Question 3

A salon offers a package priced at ₹1000.
The following services are included:

Service	Price (₹)
Hair Cut	120
Shaving	80
Hair Colour	600
Hair Wash	50

Using the average-based proportional distribution method, calculate the payout for each service.

Answer 3

Step 1: Average service price

$$
AVG = \frac{120 + 80 + 600 + 50}{4}
= \frac{850}{4}
= 212.5
$$

Step 2: Split services

- Low services: Hair Cut (120), Shaving (80), Hair Wash (50)
- High services: Hair Colour (600)

Step 3: Pay low services fully

$$
LowTotal = 120 + 80 + 50 = 250
$$

$$
R = 1000 - 250 = 750
$$

Step 4: Proportional reduction

Only one service is above average:

$$
HairColourPayout = 750
$$

Final payout

Service	Amount (₹)
Hair Cut	120
Shaving	80
Hair Wash	50
Hair Colour	750
Total	**1000**